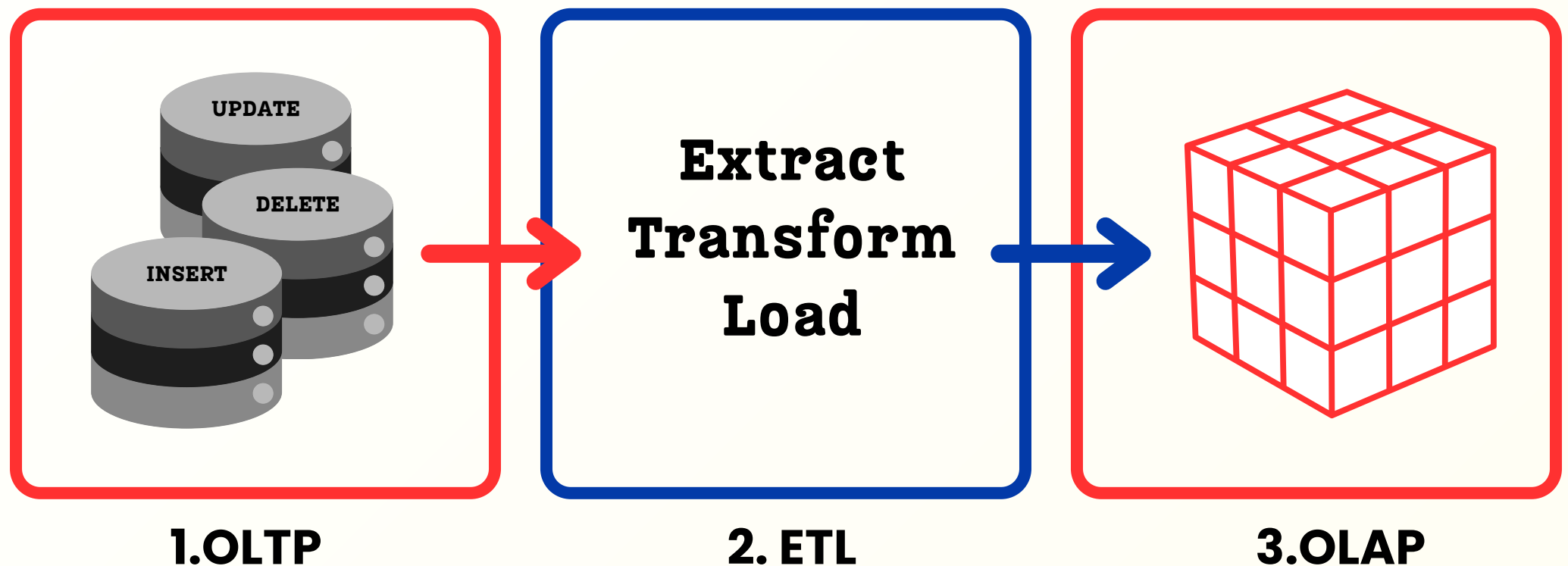


OLAP vs OLTP

Understand the difference between these two database systems and their use cases



Source: <https://medium.com/@HalderNilimesh/comprehensive-comparison-of-oltp-and-olap-key-differences-use-cases-and-choosing-the-right-565e74be4670>



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What is OLTP?

OLTP (Online Transaction Processing)

Purpose: Handles day-to-day operations like transactions.

Features:

- Fast and frequent read/write operations.
- Stores detailed and current data.
- Supports many concurrent users.

Use Case: Banking systems, retail point-of-sale systems.



What is OLAP?

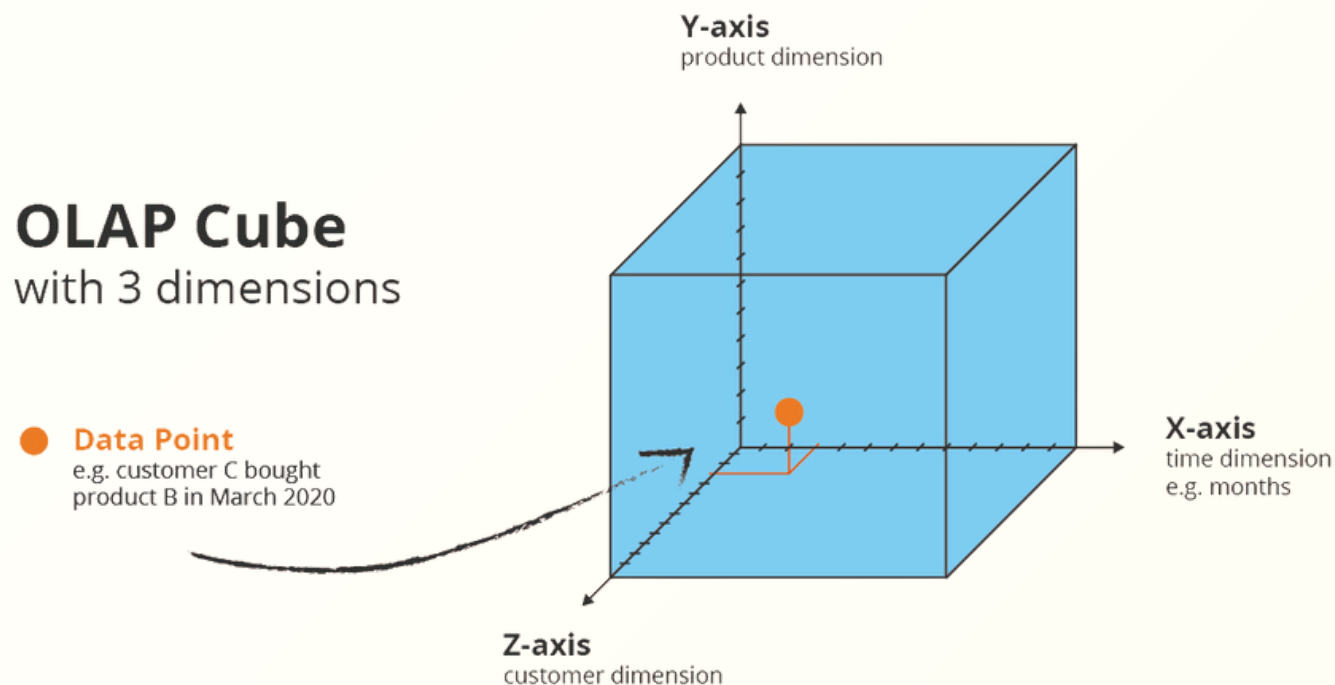
OLAP (Online Analytical Processing)

Purpose: Analyzes historical data for insights.

Features:

- Optimized for complex queries and aggregations.
- Stores summarized and historical data.
- Supports fewer, analytical users.

Use Case: Business intelligence, data warehousing.



(source: <https://www.jedox.com/en/blog/what-is-olap/>)



Key Differences

Feature	OLTP	OLAP
Purpose	Process transactions	Analyze data
Data	Current & detailed	Historical & summarized
Operations	Insert, Update, Delete	Complex queries, Aggregations
Users	Many concurrent users	Few analytical users
Performance	High for small operations	High for large queries



Real-World Examples

- **OLTP:**

- Bank transactions.
- Online shopping carts.
- Flight booking systems.

- **OLAP:**

- Quarterly sales analysis.
- Customer segmentation reports.
- Market trend forecasting.

When to Use OLTP vs OLAP?

- Use **OLTP** for operations that require **real-time transaction processing**.
- Use **OLAP** for systems that require **insights and decision-making**.

Tip: They often complement each other—OLTP systems generate data, while OLAP systems analyze it!

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